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Diffusional aging at water/oil interfaces laden with charged nanoparticles studied by single-molecule tracking

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ABSTRACT
The adsorption of charged nanoparticles at water-oil interfaces constitutes a fundamental phenomenon, underlying pivotal technologies spanning from emulsion stabilization to the sophisticated fabrication of foams. However, the diffusional behavior of these nanoparticles remains poorly understood. Here, we use single-molecule tracking experiments to show that the diffusion of like-charged nanoparticles at the water-oil interface not only becomes anomalous but also displays a diffusional aging phenomenon at the interfacial coverage that is not associated with the glassy state. We further develop a theoretical framework that quantitatively reproduces all experimental observations. Molecular dynamics simulations reveal the necessity of the coexistence of attraction and repulsion for the emergence of aging dynamics. The interplay between attraction and repulsion leads to nanoparticle adhesion taking place over observable timescales, which is manifested as aging dynamics. The discovery of diffusional aging demonstrates that interfacial evolution persists beyond adsorption equilibrium, suggesting that this effect must be accounted for in applications involving water-oil interfaces laden with charged nanoparticles.

Introduction

Microparticles and nanoparticles are capable of adsorbing at liquid/liquid interfaces by reducing interfacial energy when positioned between immiscible phases (such as water and oil)[1, 2]. This fundamental phenomenon underpins diverse applications, such as ore purification, Pickering emulsions, encapsulation, and two-dimensional film preparation[3, 4, 5]. When a single particle is adsorbed at the liquid interface, it cannot be simply regarded as a model system confined to a two-dimensional system. On the contrary, it exhibits many characteristics. For example, particles display physical aging of the contact line[6]. Moreover, their in-plane diffusion is slower than that from the prediction of the Stokes-Einstein relation[7, 8, 9, 10, 11, 12]. As the interfacial coverage of particles rises, interfacial jamming leads to the formation of a load-bearing network that transforms the system from a liquid-like state to a solid-like state[13, 14, 15]. Real-time imaging studies revealed heterogeneous dynamics during jamming, where nanoparticles exhibited spatially non-uniform organization and aging dynamics[16, 17, 18]. The underlying mechanism of this aging phenomenon lies in the relaxation processes and structural reorganization within the material, sharing characteristics with glassy systems[17, 19].

When particles carry electric charges, their interactions become substantially more complex[20, 21]. Specifically, microparticles with the same charge can be frozen into ordered two-dimensional arrays, and this can happen without the requirement for an enclosed space. This interesting phenomenon arises from the concurrent action of interparticle electrostatic repulsion and electric-field-induced capillary attraction[22, 23, 24, 25]. Compared to colloidal microparticles, nanoparticles display relatively weaker in-plane interactions[26]. As particle size shrinks to the nanoscale, thermal energy becomes sufficient to surmount electrostatic repulsive interactions. Consequently, this disrupts the ordered arrangements and allows for diffusion to occur. In bulk, this transition may lead to the formation of a 'Wigner glass' state[27, 28, 29, 30], which exhibits aging dynamics stemming from electrostatic repulsion. At water/oil interfaces, the simultaneous presence of both repulsive and attractive forces makes the diffusional behavior at liquid interfaces populated with charged nanoparticles extremely complex and challenging to predict[31].

In this work, we combined single-molecule tracking (SMT), machine learning (ML), theoretical derivation, and molecular dynamics simulation to study the diffusion at water/polydimethylsiloxane (PDMS) interfaces. These interfaces were populated with negatively charged silica nanoparticles, and the investigation was carried out after the interfacial coverage (Φ) had reached a steady state. We found that when the Φ of these particles exceeded a certain threshold, while still being far from reaching the glassy state [32, 33], the diffusion of charged nanoparticles exhibited an unexpected aging phenomenon. Through a stepwise approach combining SMT experiments, ML classification, theoretical modelling, and molecular dynamics simulations, we elucidated the underlying mechanism.

Results

SMT of charged nanoparticles at the water/PDMS interface

We first experimentally investigate the diffusion behavior of fluorescent silica nanoparticles among non-fluorescent silica negatively charged nanoparticles with varying Φ. Both types of nanoparticles possess comparable hydrodynamic radii of ≈ 70 nm (Supplementary Fig. 1a,b and Methods). They were functionalized with amphiphilic poly(ethylene glycol) (PEG, $M_n \approx 500$ g mol^{-1}) to enhance their surface activity at the water/PDMS interfaces[34, 35, 36]. Low-molecular-weight PEG was chosen to minimize the interparticle entanglements[37, 38, 39]. PEGylation led to an increase in the zeta potential from −45.6 to −23.7 mV (Supplementary Fig. 1c). The water/PDMS interface was established by depositing 5 μL of PDMS (with a viscosity of 10 mPa s) onto a clean coverslip. Subsequently, it was stabilized by positioning it within a nickel transmission electron microscopy grid. Afterwards, an aqueous solution was gently poured onto the PDMS surface. This solution contained a small amount of fluorescent nanoparticles and varying amounts of non-fluorescent nanoparticles (0-30 mg mL^{-1}). Once the water-PDMS interface had formed, a steady Φ was obtained within 30 min, as indicated by the plateau in interfacial tension (Supplementary Fig. 2). In each experiment, the concentration ratio of fluorescent nanoparticles to non-fluorescent nanoparticles was predefined. Leveraging this known parameter, the Φ of non-fluorescent nanoparticles can be calculated by counting the number of fluorescent ones. Without the addition of NaCl, the maximum surface coverage achieved in this work is 7%. This restriction arises because PEGylated silica nanoparticles carry a strong negative charge (zeta potential = −23.7 mV), generating long-range repulsion that limits surface coverage of nanoparticles at a water-oil interface (Supplementary Fig. 3, Supplementary Table 1, and Supplementary Note 1)[31]. The nanoparticles in the aqueous solution were carefully replaced with deionized water (Supplementary Fig. 4 and Methods). After this replacement, Φ showed no statistically significant variation, suggesting irreversible particle adsorption[40].

Total internal reflection fluorescence microscopy was used to enable SMT for investigating the interfacial diffusion dynamics at the water-PDMS interface (Fig. 1a and Methods). SMT is a powerful technique for investigating diffusion phenomena across various systems, including liquid-liquid interfaces, liquid-solid interfaces, and bulk phases[41, 42, 43, 44, 45, 46, 47]. A custom-developed algorithm was employed to extract trajectories from the microscopy movie[48]. In each experiment, we were able to extract at least 10^5 trajectories to ensure the statistical significance of the results. Upon analyzing the movies and trajectories, we realized that even at Φ = 7%, the trajectories exhibited a random-walk pattern, and no occurrence of immobilization or intermittent hopping was detected (Supplementary Fig. 5). To quantify the dynamics, we calculated the ensemble-averaged mean squared displacement (EA-MSD, $\langle \Delta \mathbf{r}(\tau)^2 \rangle$) and determined the values of the ensemble-averaged diffusion coefficient (D_e) and exponent α_e through $\langle \Delta \mathbf{r}(\tau)^2 \rangle = 2nD_e\tau^{\alpha_e}$ (Supplementary Note 2), where n denotes the dimension and τ is the lag time. In this analysis, n = 2 accounts for the two-dimensional motion at the liquid-liquid interface. As Φ increased, D_e decreased, and α_e gradually fell below 1, indicating the emergence of subdiffusion (Fig. 1b, Supplementary Fig. 6a, and Supplementary Table 2). Moreover, the ensemble-averaged displacement probability distribution, $P(\Delta X, \tau)$, which describes the probability density of an object moving a distance ΔX along the x- or y-axis over τ (Supplementary Note 2), transitions gradually from a Gaussian distribution to an exponentially decaying function from Φ = 0% to 7% (Fig. 1c and Supplementary Fig. 6b). This behavior was statistically validated by Kolmogorov-Smirnov tests (Supplementary Note 3). Further analysis revealed that the diffusion for Φ = 7% also displayed negatively correlated velocities and weak ergodicity-breaking behavior (Fig. 1d and Supplementary Note 2)[48].

Notably, we observed that diffusion changed over time by calculating the ensemble average of time-averaged MSD (<TA-MSD>, $\langle\overline{\delta^2(\tau,T)}\rangle$) where T denotes the temporal interval extending from the first recorded position of a specified trajectory (Supplementary Note 2). In the dilute limit (Φ = 0% and 0.09%), <TA-MSD> was independent of T, as expected (Supplementary Fig. 7a). As Φ increased, aging phenomena became evident in the form of $\overline{<\delta^2(\tau,T)>} \propto T^{\beta-1}$. At a maximum Φ of 7%, a β-1 value of −0.13 ± 0.05 was obtained (Fig. 1e and Supplementary Table 3). The diffusion dynamics also exhibits apparent non-ergodic features associated with the ageing regime (Supplementary Fig. 11a and Supplementary Note 2)[49, 50]. To verify the reliability of our observations, the ensemble average of the time-averaged diffusion coefficient $D_{T\mathrm{m}}$ was calculated at different sampling time points T_m (Supplementary Note 2). Once again, $D_{T\mathrm{m}}$ remained constant at Φ = 0% and 0.09% (Supplementary Fig. 7b), while aging phenomena gradually emerged as Φ increases (Fig. 1f). It is notable that these results did not account for the aforementioned 30-min period to exclude the aging phenomena that might originate from changes in Φ. In fact, diffusion undergoes aging shortly after adsorption occurs, even when the results from the first 30 min are included (Supplementary Fig. 8). Subsequently, the nanoparticle solution was replaced with pure water, thereby confirming that the subsequent aging process is not attributed to the rise in Φ caused by the adsorption of new nanoparticles.

When 10^{-1} M NaCl was added to the aqueous solution, the Φ increased to ≈20% (Supplementary Fig. 9a), and diffusional aging was eliminated. As shown in Fig. 1g, as the concentration of NaCl increased, <TA-MSD> decreased, and the absolute value of β-1 gradually approached zero. Because the addition of salt effectively screens electrostatic interactions[51, 52], the emergence of aging-related phenomena must be inherently linked to the electrostatic repulsion among the charged nanoparticles. Moreover, when 10^{-1} M NaCl was added to the aqueous solution, the <TA-MSD> values decreased by approximately one order of magnitude compared to those at the interface when pure water was used as the aqueous phase with an ionic concentration of ≈ 10^{-7} M, as shown in Supplementary Fig. 9b. This significant decrease cannot be solely ascribed to confinement or crowding-associated effects. The most reasonable explanation is the formation of particle aggregation[53]. Considering that the highest Φ studied was 7%, this implies the presence of attractive interactions among the nanoparticles. This hypothesis aligns with previous studies that have reported attractive capillary forces between charged particles at the water-oil and water-air interfaces[22, 23].

To verify the mechanism proposed earlier, we carried out SMT experiments that were judiciously designed to investigate the diffusion of nanoparticles at a volume fraction of 7% in the bulk. It is a well-known fact that diffusion, in this context, does not involve any attractive capillary interactions. In this experiment, we did not detect diffusional aging (Supplementary Fig. 10a,b). This result further emphasized that the attractive interactions among nanoparticles play a crucial role in initiating the diffusional aging phenomenon.

To further investigate whether nanoparticle replacement plays a role, we also conducted experiments with particles present in the bulk at Φ = 7%. As expected, D_e decreased over time (Supplementary Fig. 10c). The experimental observations are qualitatively consistent with those undergoing nanoparticle replacement (Fig. 1f).

Prediction of the anomalous diffusion model

To determine the underlying anomalous diffusion model governing interfacial diffusion, we developed a ML framework[54] that integrates feature extraction with a random-forest algorithm[50, 55, 56, 57, 58, 59, 60, 61]. This enabled us to classify these trajectories into six candidate models: annealed transient-time motion (ATTM), continuous-time random walk (CTRW), Levy walk (LW), fractional Brownian motion (FBM), scaled Brownian motion (SBM), and Brownian walk (BW) (Fig. 2a). Note that the ATTM used is a modified version with an exponential distribution of diffusion coefficient. The physical interpretations of the different diffusion models are detailed in Supplementary Note 4.

For the simulated unseen trajectories, the trained ML algorithm achieved an overall 95% accuracy. Even in the worst case of identifying FBM trajectories, the algorithm achieved a precision of 91%, a recall of 88% and an F1-score of 89%, respectively (Fig. 2b and Supplementary Note 5).

Using the trained algorithm, we found that at Φ = 0%, 85% of the trajectories were classified as BW (Fig. 2c); the fact that the classification did not reach 100% is attributed to deviations from standard BW exhibited by individual nanoparticles diffusing at the oil-water interface[62]. As Φ increases from 0.15% to 1%, the proportion of trajectories classified as BW decreases from 40% to 34%, and the proportion classified as ATTM increases from 35% to 60% (Fig. 2d-f). At Φ = 7%, the algorithm classified 62%, 4%, 9%, 5% and 20% of the trajectories as ATTM, CTRW, FBM, BW, and SBM, respectively (Fig. 2g). Although the algorithm did not identify a single dominant diffusion model, the top three models accounted for 91% of the classifications. This suggests that the diffusion behavior of the trajectories likely aligns with the features of these three diffusion models. For instance, although only 9% and 20% of the trajectories were classified as consistent with FBM and SBM, respectively, we have observed anti-correlation and aging in the diffusion process — features characteristic of these two models. Therefore, it is reasonable to hypothesize that the highest prevalence of the ATTM model may be attributed to the heterogeneity in the trajectories.

Analysis of heterogeneity in trajectories. We analyzed the heterogeneity of trajectories. Applying a change-point detection method indicated that there are no transitions in the diffusive states within individual trajectories (Fig. 3a and Supplementary Note 6). This differs from the diffusion behavior observed at the solid-liquid interface, where desorption-mediated intermittent hopping was reported[63, 64]. To characterize the heterogeneity among trajectories, the displacements within each trajectory are normalized by the standard deviation of that particular trajectory[65]. The distribution of normalized displacement $P(\Delta X_{\delta}, \tau)$ can be approximated by a Gaussian distribution (Fig. 3b). This indicates that the non-Gaussianity of $P(\Delta X, \tau)$ shown in Fig. 1c stems from heterogeneity among different trajectories.

Furthermore, by fitting the exponential tail of $P(\Delta X, \tau)$ across different τ to the functional form $P(\Delta X, \tau) \propto \exp[-|x|/\lambda(\tau)]$. The extracted length scale $\lambda(\tau)$ exhibits a scaling behavior $\lambda(\tau) \propto \tau^{1/2}$. The 1/2 exponent serves as a characteristic signature confirming the presence of multiple Gaussian states[66, 67] (Supplementary Fig. 11b, c). Taken together, the analysis unequivocally reveals pronounced trajectory-to-trajectory heterogeneity in the system[68], which is different from the heterogeneities near the colloid-gel transition[69, 70].

Physical rationale of diffusional aging

Based on the discovered features of anomalous diffusion and its aging characteristics， we developed a modified overdamped Langevin equation[71] incorporating a time-dependent diffusion coefficient, fractional Gaussian noise, and heterogeneously distributed diffusion coefficients from experimental measurements (Supplementary Note 7 and Supplementary Fig. 12). Numerical simulations using this model showed excellent agreement with experimental data, confirming its effectiveness in capturing the complex diffusion behavior at water-PDMS interfaces (Fig. 1b-e). The observed anomalous diffusion and its aging characteristics bear a resemblance to the glassy dynamics seen in repulsive colloidal systems, such as Wigner glass[30]. However, it is important to note that glassy dynamics do not exhibit multiple Gaussian states.

We propose that diffusional aging arises from the coexistence of long-range electrostatic repulsion and short-range capillary attraction. A key experimental support for this scenario is that if we screen the repulsive interactions by adding salt, we observed an increase in surface coverage and a decrease in the diffusion coefficient, but no aging phenomenon occurred (Supplementary Figs. 1g, 9). In this scenario, the decrease in D_{Tm} can be interpreted as a result of particle aggregation. This indicates that repulsive interactions play a decisive role in the emergence of diffusional aging and suggests the presence of attractive interactions between particles. In addition to the intrinsic interactions between particles, capillary forces also arise in the context of particles at water-oil interfaces, which can further drive particle aggregation. Therefore, diffusional aging can be understood as a result of particles occasionally overcoming the long-range electrostatic repulsion that separates them. When they approach closely enough, short-range attractive interactions, e.g., the capillary forces, become dominant, leading to aggregation. Although individual particle size remains unchanged, these aggregation events progressively and slowly increase the size of the diffusion species over time. This phenomenon corresponds to a scaling behavior $\lambda(\tau) \propto \tau^{1/2}$ (Supplementary Fig. 11), again indicating that the diffusion of each aggregate remains Gaussian. This behavior is markedly distinct from conventional aggregation driven by a balance of enthalpic attractions and entropic forces, which typically proceeds rapidly toward thermodynamic equilibrium[72, 73].

Molecular dynamics of 2D particle diffusion

To elucidate the underlying mechanisms governing the observed aging dynamics and anomalous diffusion, we further performed molecular dynamics simulations[74] of charged particle diffusion in two dimensions. The interparticle interactions were modelled using a long-range repulsive Yukawa potential to account for electrostatic repulsion, combined with a short-range generalized Lennard-Jones (LJ) attractive potential to capture electric-field-induced capillary attraction (Supplementary Fig. 13). We simulated particle diffusion at Φ = 0.1%-10% and a reduced temperature of θ = 0.1. All quantities are described in standard reduced LJ units; e.g., length, time, and temperature are in units of σ, $\sqrt{m\sigma^2/\varepsilon}$, and $\varepsilon/\mathrm{k_B}$, where σ and ε are the length and energy scales associated with the LJ potential, m is the particle mass, and $\mathrm{k_B}$ is Boltzmann's constant. All molecular dynamics simulations were conducted in the canonical ensemble (*NVT*).

Here, we present results obtained at an area fraction of Φ = 7%. With increasing time, the system evolves from a state of individual, well-dispersed particles to the formation of aggregates (Fig. 4a,b). Quantitative analysis demonstrates that the number of nanoparticles per aggregate increases over time (Fig. 4c and Supplementary Fig. 14), accompanied by a growing heterogeneity in the number distribution of nanoparticles within aggregates. Meanwhile, the diffusion coefficient is found to decrease progressively over time (Fig. 4d). Importantly, when Φ is reduced to 1%, almost no large aggregate formation is observed (Fig. 4e-g), and the diffusion coefficient remains essentially time-invariant (Fig. 4h). This suggests that the slow emergence of heterogeneous aggregates is responsible for the aging behavior under higher Φ conditions.

To gain a deeper understanding of the physical mechanisms behind slow aggregation, we carried out control simulations by regulating the parameter α that controls the range of attractive interaction at Φ = 1% and a temperature of 0.1 (Supplementary Fig. 15). Our findings indicate that the range of attractive interactions is also essential for the emergence of aging behavior. At α = 100, where the attraction range is very short, almost no aggregates are formed within the time window of our simulations, and no signatures of aging dynamics are observed. However, as α decreases, aggregate formation becomes apparent, and the diffusion coefficient exhibits a clear time-dependent decay (Supplementary Figs. 16, 17).

Validation of progressive nanoparticle aggregate growth

To visualize nanoparticles at the oil-water interface, we employed a gel-trapping technique[75, 76] (Fig. 5a and Methods). To this end, nanoparticles were adsorbed at an interface between octane and a 1, 1.5, and 2 wt.% gellan gum aqueous solution at 50 °C. Upon cooling at 25 °C, the gellan gum underwent gelation, thereby fixing the nanoparticle positions at different times after adsorption equilibrium. After removal of the octane phase, liquid PDMS prepolymer was poured over the gel surface, cured, and subsequently peeled off to produce a replica of the particle dispersion state for scanning electron microscopy observation (Fig. 5b-d).

As time progresses, we observe the nanoparticle dispersion state evolving from predominantly individual particles and small aggregates composed of only a few nanoparticles, to larger aggregates exhibiting increasing heterogeneity in nanoparticle number. Consequently, we applied a counting technique to determine the exact number of nanoparticles within individual aggregates at each time point (Supplementary Fig. 18 and Supplementary Note 8) and plotted these data as a function of time (Fig. 5e,f). The results confirm our observation that the number of nanoparticles per aggregate increases over time, with a correspondingly broader and more heterogeneous distribution. This temporal evolution aligns qualitatively with the physical mechanism we proposed, and the aggregate growth behavior observed in the simulations. Calculating the inter-nanoparticle potential helps understand their interactions, but sufficient particle observation is required. We used high-magnification imaging (22000×, with over 5000 particles) for clear aggregate images. However, the particle number obtained is still below the statistical threshold, and variations in particle detection accuracy across different SEM images further compromise data reliability, making quantitative inter-nanoparticle potential calculation unfeasible.

Discussion

SMT experiments have demonstrated that diffusional aging occurs during the diffusion of charged nanoparticles at water/PDMS interfaces when Φ exceeds 0.15%. This is accompanied by multiple characteristics such as subdiffusion, negatively correlated velocities, and a non-Gaussian $P(\Delta X, \tau)$. The ML-assisted trajectory classification algorithm revealed that the diffusion did not adhere to any preexisting theoretical models; however, they did share certain key characteristics with FBM, SBM, and ATTM. To model the observed diffusion behavior, we proposed a modified overdamped Langevin equation incorporating fractional derivatives and a time-dependent diffusivity $D(t)$ that was distributed exponentially. Numerical simulations using the proposed model exhibited quantitative agreement with the experimental data. To elucidate the underlying mechanisms, we conducted molecular dynamics simulations that revealed the necessity of the coexistence of attraction and repulsion for the emergence of aging dynamics. The interplay between attraction and repulsion leads to aggregation taking place slowly over observable timescales, which is manifested as aging dynamics. The discovery of diffusional aging reveals that interfacial evolution persists even after adsorption equilibrium is reached. The discovery of this phenomenon offers a mechanistic understanding of charged nanoparticle interfacial dynamics, with particular relevance to applications involving charged nanoparticles at liquid interfaces.

Methods

Materials

Ethanol (EtOH, ⩾99.8%), (3-aminopropyl) triethoxysilane (APTES, 97%), tetraethyl orthosilicate (TEOS, 98%), and ammonium hydroxide (AR, 25%-28%) were purchased from Aladdin Reagent. Rhodamine B isothiocyanate (RhB-ITC) was purchased from Sigma-Aldrich. 2-[methoxy-(polyethyleneoxy)propyl]trimethoxysilane (PEG-silane, $M_n \approx 500$ g mol^{-1}) was purchased from J&K Scientific. Polydimethylsiloxane (PDMS, viscosity = 10 mPa s at 25 °C), gellan gum, n-octane, basic alumina, acetonitrile, and isopropanol were purchased from Macklin Biochemical Technology. C18-silica chromatographic columns were purchased from Delvstlab, and PDMS Sylgard 184 elastomer was supplied by Dow Corning. Water was purified using a Milli-Q system, resulting in a final resistivity of 18 MΩ cm.

Synthesis of fluorescent PEGylated silica nanoparticles

RhB-ITC in ethanol was conjugated to APTES before silica shell synthesis in a molar ratio of 1:25, ensuring covalent binding to the RhB-ITC molecules. The solution was vigorously stirred for 24 h under dark conditions to facilitate the formation of the RhB-APTES conjugate.

To synthesize the fluorescent PEGylated silica nanoparticles (F-PEG-NPs), aqueous ammonia (0.88 mL), water (0.15 mL), and ethanol (25 mL) were mixed and stirred at room temperature. After allowing the mixture to equilibrate for 10 min, 125 μL of the RhB-APTES conjugate was added and vigorously stirred at room temperature overnight. Subsequently, five separate additions of 40 μL of TEOS were carefully added drop by drop at 1-h intervals to facilitate the continued attachment of RhB-APTES to the silica nanoparticles. The mixing rate was reduced to 150 rpm, resulting in highly dye-doped particles. Following 7-8 h of growth, 250 μL PEG-silane was introduced, and the solution was vigorously stirred (1000 rpm) once again at room temperature for 12 h.

Afterwards, the solution was heated to 80 °C, and stirring was halted for a minimum of 12 h. After cooling, the F-PEG-NPs were isolated through centrifugation at 11180 × *g* for 10 min and consecutively washed with ethanol and water. Ultimately, the F-PEG-NPs were dried under vacuum conditions for 12 h.

Synthesis of non-fluorescent PEGylated silica nanoparticles

An ammonium solution (8.75 mL, 30%−33%) and 25 mL of ethanol were added to a 25-mL round-bottomed flask. After equilibration for 10 min, 20 mL of TEOS was quickly added to the mixture. The reaction proceeded for 24 h at room temperature. The resulting mixture was separated by centrifugation at 11180 × *g* for 20 min and successively washed with ethanol and water. Finally, the silica nanoparticles were dried under vacuum for 12 h.

Subsequently, the silica nanoparticles (100 mg) were dispersed in 25 mL of water by ultrasonication. The solution's pH was adjusted to 9-10 using small amounts of ammonium hydroxide solution. Next, 500 µL of PEG-silane was added, following the same procedure as the synthesis of F-PEG-NPs. The purification steps were identical to those previously outlined in the preparation of F-PEG-NPs.

Characterization of nanoparticle size and zeta potential

The hydrodynamic diameter and zeta potential of the synthesized nanoparticles were determined using dynamic light scattering (DLS) and zeta potential measurements, respectively. These measurements were performed on a Malvern Zetasizer Nano ZS (Malvern, UK), employing a 633-nm HeNe laser. Before analysis, the samples (≈1 mg mL^{-1}) were dispersed in the water and filtered through a 220 nm syringe filter (Millex) to remove any dust and aggregation. The reported size and zeta potential values represent the average of three independent measurements.

The charged non-fluorescent PEGylated silica nanoparticles (PEG-NPs) and fluorescent PEGylated silica nanoparticles (F-PEG-NPs) were synthesized according to established protocols[77, 78]. The PEG-NPs were synthesized using the classic Stöber method, involving the hydrolysis-condensation reaction of tetraethyl orthosilicate (TEOS). The F-PEG-NPs consisted of a Rhodamine B-(3-aminopropyl) triethoxysilane (RhB-APTES) conjugate core, encapsulated by a silica shell formed via TEOS hydrolysis-condensation. Both PEG-NPs and F-PEG-NPs were subsequently functionalized with 2-[methoxy-(polyethyleneoxy)propyl]trimethoxysilane (PEG-silane) to ensure surface PEGylation.

Sample preparation for single-molecule tracking

The water solution contained varying concentrations of PEG-NPs and adjustable concentrations of F-PEG-NPs, ranging from 10^{-11} to 10^{-13} M. The concentration of F-PEG-NPs was adjusted proportionally to the concentration of PEG-NPs to maintain a constant mean interfacial coverage of fluorescent probes at ≈0.01 particle per μm^2. Before imaging, a solution exchange operation was performed to remove unadsorbed NPs. The specific procedure was as follows: after allowing at least 30 min for the water/oil interface to stabilize and reach nanoparticle adsorption equilibrium, 500 µL of the PEG-NPs aqueous solution was carefully extracted using a pipette from the total 1.2 mL of nanoparticle solution.

Subsequently, an equal volume (500 µL) of pure water (without nanoparticles) was added to the sample cell. To significantly reduce the nanoparticle concentration in the bulk solution while minimizing disturbance to the nanoparticles already adsorbed at the interface, and enhance the reproducibility and controllability of the subsequent experiments, this exchange procedure was repeated five times before conducting the subsequent single-molecule fluorescence experiments. This step was followed by a stabilization period of about 30 min to allow the water/oil interface to equilibrate and reach nanoparticle adsorption equilibrium.

Total internal reflection fluorescence microscopy

We performed single-molecule tracking using a total internal reflection fluorescence microscope. The system was based on a Ti2-E inverted microscope, equipped with a high-numerical-aperture 100× oil-immersion objective, a laser beam deflection module, and an electron-multiplying charge-coupled device camera (iXon DU897), operated at −70 °C to capture image sequences. F-PEG-NPs were excited using a continuous-wave 561 nm OBIS-LX diode laser. TIRFM utilizes the evanescent field generated by total internal reflection of the laser at a liquid–liquid interface to selectively excite F-PEG-NPs located within ≈ 100 nm of the interface. The refractive indices of the objective oil, glass slide, water, and PDMS were 1.52, 1.52, 1.33, and 1.43, respectively. To achieve total internal reflection at the water/PDMS interface, the incident angle of the laser was adjusted to be ≈70°. It should be noted that only nanoparticles adsorbed at the interface, experiencing the viscosity of PDMS, could be identified, as the rapid diffusion of nanoparticles in bulk water exceeds the resolution limit of the microscope setup. All measurements were carried out at room temperature.

Data acquisition and analysis

For each Φ, more than 30 movies, each lasting 1 min, were continuously captured at multiple locations and repeated on multiple days to ensure reliability and reproducibility. The centroid-of-intensity positions of nanoparticles were identified in each image with a spatial resolution of 10-20 nm, and trajectories were constructed from the image sequences using a custom-developed tracking algorithm[48]. All statistical analyses were rigorously performed on a dataset of at least 1000 trajectories, each exceeding a duration of 2.1 s.

Machine learning for diffusion model classification

To improve the interpretability of the results, we chose the random forest algorithm — which employs decision trees as base learners — as our classification method[79, 80]. To enhance the performance of the classifier, we extended the original feature framework described in prior work[57] by incorporating additional analyses, such as power spectral density (PSD)[81, 82]. For training, we generated 10000 trajectories (each of length 100) from six standard models: ATTM, CTRW, FBM, LW, SBM, and BW. To better mimic experimental conditions, each trajectory was intentionally perturbed by introducing a finite localization error. To simulate this, random noise sampled from a normal distribution Normal(0, σ_{noise} = 0.1) was added to each coordinate of the trajectories. The whole dataset was then partitioned into training (70%), validation (15%), and test subsets (15%).

Molecular dynamics simulation

We simulated a 2D system of N = 1000 particles in a periodic box with an area fraction $\Phi = \pi\sigma^2 N/(4S)$, where σ is the particle diameter and S is the box area, using established methods[83, 84]. Periodic boundary conditions were applied along both x and y directions. Pairwise interactions combined a short-range Lennard-Jones attraction whose exponent α can be varied to tune the range of attractive interactions and a long-range Yukawa repulsion ($r_{\text{cut}} = 8\sigma$), with parameters $\xi = 2\sigma$ (Debye screening length) and $A = 0.2\varepsilon$ (Coulombic prefactor),

$$U(\mathrm{r}) = 4\varepsilon\left[\left(\frac{\sigma}{r}\right)^{2\alpha} - \left(\frac{\sigma}{r}\right)^{\alpha}\right] + \frac{A\xi}{r}\mathrm{e}^{-r/\xi} \quad (1)$$

Molecular dynamics simulations (*NVT* ensemble) were performed with the Large-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS) package (version June 23, 2022)[85, 86], integrating the equations of motion with a time step δ_t = 0.001 and maintaining the temperature via a Nosé-Hoover thermostat.

Gel trapping technique

To visualize nanoparticles at the water/oil interface, we employed the gel-trapping technique following the reported protocol[75, 76]. A gellan dispersion in Milli-Q water was heated to 95 °C for 15 min, clarified by two passes through a C18 column pre-activated with acetonitrile–water (v/v = 80:20) and rinsed with hot Milli-Q water, lyophilised, and redissolved to 1-2 wt.% using the same heating protocol. The hot solution was held at 50 °C, poured into a small petri dish, and overlaid with n-octane purified three times over basic alumina. A 10 µL suspension of 1 wt.% monodisperse PEG-SiO_2 in isopropanol was deposited on the hot interface; the dish was immediately lidded, cooled to ≈ 25 °C, and left to allow both gelation and evaporation of the octane. PDMS (Sylgard 184, 10:1 base: cure) was centrifugally degassed at 224 × *g*, gently cast on the gelled surface, and cured at 25 °C for 48 h. After peeling off the solid elastomer, the particles were imprinted onto the PDMS. Residual gellan was removed by immersing the sample in 95 °C Milli-Q water for 5 min, followed by thorough rinsing with Milli-Q water. The sample was gently dried with N_2 flow and then observed by scanning electron microscopy (SEM, SU-8600, Hitachi) after Au sputter-coating.

Data availability

The data that support the findings of this study are available from the corresponding authors upon request.

Unprocessed SEM images are provided as Supplementary Data 1. Source data are provided with this paper.

Code availability

Custom codes used in this work, including AI training scripts[54], numerical simulation routines[71], and LAMMPS input files[74], are available from Figshare, and from the corresponding authors upon request. These codes are released under the MIT open-source license. Comprehensive documentation for each module is included in the archives.

References

1. Bresme, F. & Oettel, M. Nanoparticles at fluid interfaces. *J. Phys.: Condens. Matter* **19**, 413101 (2007).
2. Garbin, V., Crocker, J.C. & Stebe, K.J. Nanoparticles at fluid interfaces: exploiting capping ligands to control adsorption, stability and dynamics. *J. Colloid Interface Sci.* **387**, 1-11 (2012).
3. Shi, S. & Russell, T.P. Nanoparticle assembly at liquid-liquid interfaces: From the nanoscale to mesoscale. *Adv. Mater.* **30**, e1800714 (2018).
4. Binks, B.P. Colloidal Particles at a range of fluid-fluid interfaces. *Langmuir* **33**, 6947-6963 (2017).
5. Guzman, E. et al. A broad perspective to particle-laden fluid interfaces systems: from chemically homogeneous particles to active colloids. *Adv. Colloid Interface Sci.* **302**, 102620 (2022).
6. Kaz, D.M., McGorty, R., Mani, M., Brenner, M.P. & Manoharan, V.N. Physical ageing of the contact line on colloidal particles at liquid interfaces. *Nat. Mater.* **11**, 138-142 (2012).
7. Wang, D. et al. Probing diffusion of single nanoparticles at water-oil interfaces. *Small* **7**, 3502-3507 (2011).
8. Wang, D. et al. Brownian diffusion of individual Janus nanoparticles at water/oil interfaces. *ACS Nano* **14**, 10095-10103 (2020).
9. Gehring, T. & Fischer, T.M. Diffusion of nanoparticles at an air/water interface is not invariant under a reversal of the particle charge. *J. Phys. Chem. C* **115**, 23677-23681 (2011).
10. Zhong, Y., Zhao, L., Tyrlik, P.M. & Wang, G. Investigating diffusing on highly curved water–oil interface using three-dimensional single particle tracking. *J. Phys. Chem. C* **121**, 8023-8032 (2017).
11. Boniello, G. et al. Brownian diffusion of a partially wetted colloid. *Nat. Mater.* **14**, 908-911 (2015).
12. Stocco, A., Mokhtari, T., Haseloff, G., Erbe, A. & Sigel, R. Evanescent-wave dynamic light scattering at an oil-water interface: Diffusion of interface-adsorbed colloids. *Phys. Rev. E* **83**, 011601 (2011).
13. Gholamipour-Shirazi, A. et al. Transition from glass-to gel-like states in clay at a liquid interface. *Sci. Rep.* **6**, 37239 (2016).
14. Thorneywork, A.L., Abbott, J.L., Aarts, D. & Dullens, R.P.A. Two-dimensional melting of colloidal hard spheres. *Phys. Rev. Lett.* **118**, 158001 (2017).
15. Liu, Z., Guo, R., Xu, G., Huang, Z. & Yan, L.T. Entropy-mediated mechanical response of the interfacial nanoparticle patterning. *Nano Lett.* **14**, 6910-6916 (2014).
16. Chai, Y. et al. Direct observation of nanoparticle-surfactant assembly and jamming at the water-oil interface. *Sci. Adv.* **6**, eabb8675 (2020).
17. Kim, P.Y. et al. Relaxation and aging of nanosphere assemblies at a water-oil interface. *ACS Nano* **16**, 8967-8973 (2022).
18. Tong, H., Hu, H., Tan, P., Xu, N. & Tanaka, H. Revealing inherent structural characteristics of jammed particulate packings. *Phys. Rev. Lett.* **122**, 215502 (2019).
19. Saltzman, E.J., Yatsenko, G. & Schweizer, K.S. Anomalous diffusion, structural relaxation and shear thinning in glassy hard sphere fluids. *J. Phys.: Condens. Matter* **20**, 244129 (2008).
20. Ridel, L., Bolzinger, M.A., Gilon-Delepine, N., Dugas, P.Y. & Chevalier, Y. Pickering emulsions stabilized by charged nanoparticles. *Soft Matter* **12**, 7564-7576 (2016).
21. Striolo, A. et al. Particles at interfaces: general discussion. *Faraday Discuss.* **191**, 407-434 (2016).
22. Nikolaides, M.G. et al. Electric-field-induced capillary attraction between like-charged particles at liquid interfaces. *Nature* **420**, 299-301 (2002).

23. Danov, K.D. & Kralchevsky, P.A. Interaction between like-charged particles at a liquid interface: electrostatic repulsion vs. electrocapillary attraction. *J. Colloid Interface Sci.* **345**, 505-514 (2010).
24. Chen, W., Tan, S.S., Ng, T.K., Ford, W.T. & Tong, P. Long-ranged attraction between charged polystyrene spheres at aqueous interfaces. *Phys. Rev. Lett.* **95**, 218301 (2005).
25. Masschaele, K., Park, B.J., Furst, E.M., Fransaer, J. & Vermant, J. Finite ion-size effects dominate the interaction between charged colloidal particles at an oil-water interface. *Phys. Rev. Lett.* **105**, 048303 (2010).
26. McDonald, M., Showunmi, S., Crimmins, M. & Tran-Ba, K.H. Single-particle diffusion at the vertical silicone oil-water interface. *ChemistrySelect* **10**, e202406040 (2025).
27. Zaccarelli, E., Andreev, S., Sciortino, F. & Reichman, D.R. Numerical investigation of glassy dynamics in low-density systems. *Phys. Rev. Lett.* **100**, 195701 (2008).
28. Courtland, R.E. & Weeks, E.R. Direct visualization of ageing in colloidal glasses. *J. Phys.: Condens. Matter* **15**, S359-S365 (2002).
29. Tanaka, H., Jabbari-Farouji, S., Meunier, J. & Bonn, D. Kinetics of ergodic-to-nonergodic transitions in charged colloidal suspensions: Aging and gelation. *Phys. Rev. E* **71**, 021402 (2005).
30. Bonn, D., Tanaka, H., Wegdam, G., Kellay, H. & Meunier, J. Aging of a colloidal "Wigner" glass. *Europhys. Lett.* **45**, 52-57 (1999).
31. Reincke, F. et al. Understanding the self-assembly of charged nanoparticles at the water/oil interface. *Phys. Chem. Chem. Phys.* **8**, 3828-3835 (2006).
32. Chaudhuri, P., Berthier, L. & Kob, W. Universal nature of particle displacements close to glass and jamming transitions. *Phys. Rev. Lett.* **99**, 060604 (2007).
33. Weeks, E.R. Introduction to the colloidal glass transition. *ACS Macro Lett.* **6**, 27-34 (2016).
34. Li, Z., Yang, J., Hollingsworth, J.V. & Zhao, J. Lateral diffusion of single polymer molecules at interfaces between water and oil. *RSC Adv.* **10**, 16565-16569 (2020).
35. Vialetto, J., Camerin, F., Ramakrishna, S.N., Zaccarelli, E. & Isa, L. Exploring the 3D conformation of hard-core soft-shell particles adsorbed at a fluid interface. *Adv. Sci.* **10**, 2303404 (2023).
36. Wang, D. et al. Scaling of Polymer Dynamics at an Oil–Water Interface in Regimes Dominated by Viscous Drag and Desorption-Mediated Flights. *J. Am. Chem. Soc.* **137**, 12312-12320 (2015).
37. Kim, P.Y. et al. Assessing pair interaction potentials of nanoparticles on liquid interfaces. *ACS Nano* **13**, 3075-3082 (2019).
38. Garbin, V., Jenkins, I., Sinno, T., Crocker, J.C. & Stebe, K.J. Interactions and stress relaxation in monolayers of soft nanoparticles at fluid-fluid interfaces. *Phys. Rev. Lett.* **114**, 108301 (2015).
39. Isa, L. et al. Core–shell nanoparticle monolayers at planar liquid–liquid interfaces: effects of polymer architecture on the interface microstructure. *Soft Matter* **9**, 3789-3797 (2013).
40. Hua, X., Bevan, M.A. & Frechette, J. Competitive adsorption between nanoparticles and surface active ions for the oil–water interface. *Langmuir* **34**, 4830-4842 (2018).
41. Shen, H. et al. Single particle tracking: from theory to biophysical applications. *Chem. Rev.* **117**, 7331-7376 (2017).
42. Higgins, D.A., Park, S.C., Tran-Ba, K.-H. & Ito, T. Single-molecule investigations of morphology and mass transport dynamics in nanostructured materials. *Annu. Rev. Anal. Chem.* **8**, 193-216 (2015).
43. Landfield, H. & Wang, M. Determination of hydrophobic polymer clustering in concentrated aqueous solutions through single-particle tracking diffusion studies. *Macromolecules* **55**, 7425-7437 (2022).
44. Ow, M.J.K. & Yeow, E.K.L. Revealing the existence of long-range liquid–liquid interfacial potential in phase-transfer processes. *J. Phys. Chem. Lett.* **15**, 6241-6248 (2024).
45. Niu, Q. & Wang, D.P. Probing the polymer anomalous dynamics at solid/liquid interfaces at the single-molecule level. *Curr. Opin. Colloid Interface Sci.* **39**, 162-172 (2019).
46. Wang, D. & Schwartz, D.K. Non-Brownian Interfacial Diffusion: Flying, Hopping, and Crawling. *J. Phys. Chem. C* **124**, 19880-19891 (2020).
47. McDonald, M., Showunmi, S., Crimmins, M. & Tran-Ba, K.-H. Single-Particle Diffusion at the Vertical Silicone Oil-Water Interface. *ChemistrySelect* **10**, e202406040 (2025).
48. Zhao, Y., Lu, Y. & Wang, D. Tracking of nanoparticle diffusion at a liquid-liquid interface adsorbed by nonionic surfactants. *Langmuir* **37**, 12118-12127 (2021).
49. He, Y., Burov, S., Metzler, R. & Barkai, E. Random Time-Scale Invariant Diffusion and Transport Coefficients. *Phys. Rev. Lett.* **101,** 058101 (2008).
50. Metzler, R., Jeon, J.-H., Cherstvy, A.G. & Barkai, E. Anomalous diffusion models and their properties: non-stationarity, non-ergodicity, and ageing at the centenary of single particle tracking. *Phys. Chem. Chem. Phys.* **16**, 24128-24164

(2014).
51. Geisel, K., Isa, L. & Richtering, W. The compressibility of pH-sensitive microgels at the oil–water interface: higher charge leads to less repulsion. *Angew. Chem. Int. Ed.* **126**, 5005-5009 (2014).
52. Chai, Y., Lukito, A., Jiang, Y., Ashby, P.D. & Russell, T.P. Fine-tuning nanoparticle packing at water-oil interfaces using ionic strength. *Nano Lett.* **17**, 6453-6457 (2017).
53. Kutuzov, S. et al. On the kinetics of nanoparticle self-assembly at liquid/liquid interfaces. *Phys. Chem. Chem. Phys.* **9**, 6351-6358 (2007).
54. Zhao, Y. Machine learning for anomalous diffusion model. *Figshare,* https://doi.org/10.6084/m9.figshare.30061978 (2026).
55. Clegg, P.S. Characterising soft matter using machine learning. *Soft Matter* **17**, 3991-4005 (2021).
56. Guo, H., Ramakrishnan, S., Harden, J.L. & Leheny, R.L. Gel formation and aging in weakly attractive nanocolloid suspensions at intermediate concentrations. *J. Chem. Phys.* **135**, 154903 (2011).
57. Kowalek, P., Loch-Olszewska, H., Łaszczuk, Ł., Opała, J. & Szwabiński, J. Boosting the performance of anomalous diffusion classifiers with the proper choice of features. *J. Phys. A: Math. Theor.* **55**, 244005 (2022).
58. Munoz-Gil, G. et al. Objective comparison of methods to decode anomalous diffusion. *Nat. Commun.* **12**, 6253 (2021).
59. Seckler, H. & Metzler, R. Bayesian deep learning for error estimation in the analysis of anomalous diffusion. *Nat. Commun.* **13**, 6717 (2022).
60. Muñoz-Gil, G. et al. Quantitative evaluation of methods to analyze motion changes in single-particle experiments. *Nat. Commun.* **16**, 6749 (2025).
61. Granik, N. et al. Single-Particle Diffusion Characterization by Deep Learning. *Biophys. J.* **117**, 185-192 (2019).
62. Wang, D., Hu, R., Skaug, M.J. & Schwartz, D.K. Temporally anticorrelated motion of nanoparticles at a liquid interface. *J. Phys. Chem. Lett.* **6**, 54-59 (2015).
63. Wang, D.P., Chin, H.Y., He, C.L., Stoykovich, M.P. & Schwartz, D.K. Polymer Surface Transport Is a Combination of in-Plane Diffusion and Desorption-Mediated Flights. *ACS Macro Lett.* **5**, 509-514 (2016).
64. Wang, D., Wu, H. & Schwartz, D.K. Three-Dimensional Tracking of Interfacial Hopping Diffusion. *Phys. Rev. Lett.* **119**, 268001 (2017).
65. Lampo, T.J., Stylianidou, S., Backlund, M.P., Wiggins, P.A. & Spakowitz, A.J. Cytoplasmic RNA-Protein Particles Exhibit Non-Gaussian Subdiffusive Behavior. *Biophys. J.* **112**, 532-542 (2017).
66. Miotto, J.M., Pigolotti, S., Chechkin, A.V. & Roldán-Vargas, S. Length scales in Brownian yet non-Gaussian dynamics. *Phys. Rev. X* **11**, 031002 (2021).
67. Hu, M. et al. Triggering Gaussian-to-Exponential Transition of Displacement Distribution in Polymer Nanocomposites via Adsorption-Induced Trapping. *ACS Nano* **17**, 21708-21718 (2023).
68. Chechkin, A.V., Seno, F., Metzler, R. & Sokolov, I.M. Brownian yet non-Gaussian diffusion: from superstatistics to subordination of diffusing diffusivities. *Phys. Rev. X* **7**, 021002 (2017).
69. Brambilla, G. et al. Probing the equilibrium dynamics of colloidal hard spheres above the mode-coupling glass transition. *Phys. Rev. Lett.* **102**, 085703 (2009).
70. Gao, Y. & Kilfoil, M.L. Direct imaging of dynamical heterogeneities near the colloid-gel transition. *Phys. Rev. Lett.* **99**, 078301 (2007).
71. Zhao, Y. Numerical simulation. *Figshare*, https://doi.org/10.6084/m9.figshare.29610338 (2026).
72. Lu, P.J. et al. Gelation of particles with short-range attraction. *Nature* **453**, 499-503 (2008).
73. Poon, W.C. The physics of a model colloid–polymer mixture. *J. Phys.: Condens. Matter* **14**, R859 (2002).
74. Zhao, Y. Molecular dynamics simulation. *Figshare*, https://doi.org/10.6084/m9.figshare.29609678 (2026).
75. Paunov, V.N. & Cayre, O.J. Supraparticles and "Janus" Particles Fabricated by Replication of Particle Monolayers at Liquid Surfaces Using a Gel Trapping Technique. *Adv. Mater.* **16**, 788-791 (2004).
76. Paunov, V.N. Novel method for determining the three-phase contact angle of colloid particles adsorbed at air− water and oil− water interfaces. *Langmuir* **19**, 7970-7976 (2003).
77. Kao, T. et al. Fluorescent Silica Nanoparticles with well-separated intensity distributions from batch reactions. *Nano Lett.* **18**, 1305-1310 (2018).
78. Stöber, W., Fink, A. & Bohn, E. Controlled growth of monodisperse silica spheres in the micron size range. *J. Colloid Interface Sci.* **26**, 62-69 (1968).
79. Muñoz-Gil, G., Garcia-March, M.A., Manzo, C., Martín-Guerrero, J.D. & Lewenstein, M. Single trajectory characterization via machine learning. *New J. Phys.* **22**, 013010 (2020).
80. Janczura, J., Kowalek, P., Loch-Olszewska, H., Szwabiński, J. & Weron, A. Classification of particle trajectories in living cells: Machine learning versus statistical testing hypothesis for fractional anomalous diffusion. *Phys. Rev. E* **102**,

032402 (2020).

81. Vilk, O. et al. Classification of anomalous diffusion in animal movement data using power spectral analysis. *J. Stat. Mech.: Theory Exp.* **55**, 334004 (2022).
82. Metzler, R. Brownian motion and beyond: first-passage, power spectrum, non-Gaussianity, and anomalous diffusion. *J. Stat. Mech.: Theory Exp.* **2019**, 114003 (2019).
83. Toledano, J.C.F., Sciortino, F. & Zaccarelli, E. Colloidal systems with competing interactions: from an arrested repulsive cluster phase to a gel. *Soft Matter* **5**, 2390-2398 (2009).
84. Mani, E., Lechner, W., Kegel, W.K. & Bolhuis, P.G. Equilibrium and non-equilibrium cluster phases in colloids with competing interactions. *Soft Matter* **10**, 4479-4486 (2014).
85. Thompson, A.P. et al. LAMMPS - a flexible simulation tool for particle-based materials modeling at the atomic, meso, and continuum scales. *Comput. Phys. Commun.* **271**, 108171 (2022).
86. Plimpton, S. Fast parallel algorithms for short-range molecular dynamics. J. Comput. Phys. 117, 1-19 (1995).

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Author contributions statement

D.W. conceived the research and coordinated the project; Y.Z. performed experiments; Y.Z., M.H., and H.C. developed the computational model; W.X. performed molecular dynamics simulations. All authors discussed the results and contributed to the writing of the paper.

Competing interests statement

The authors declare no competing interests.

Figure Legends

Fig. 1 Nanoparticle diffusion dynamics at water-PDMS interfaces for Φ = 0.15%–7%. a Schematic representation of the SMT experiments at the water-oil interface, NPs, nanoparticles; FNPs, fluorescent nanoparticles. NA, numerical aperture. **b–f**, Data for particle 0.15% (blue circle), 1% (orange diamond), and 7% (gray square). **g,** Data for NaCl concentrations of 10^{-7} M (blue circle), 10^{-4} M (orange diamond), and 10^{-1} M (gray square). **b** EA-MSD as a function of τ. **c** Displacement probability distributions $P(\Delta X,\tau)$ at τ = 0.5 s. **d** Velocity autocorrelation function $C_{\delta t}(\tau)/C_{\delta t}(0)$ (δ = 0.05 s) measured at t = 30 min (after reaching steady surface coverage) and comparison of TA-MSD, <TA-MSD>, and EA-MSD at Φ = 7%. **e** <TA-MSD> as a function of T of given trajectories at various Φ, the data were fitted to the power-law relation. Solid lines in panels **b**–**e** represent numerical simulations. **f** $D_{T\mathrm{m}}$ *vs.* T_m for various Φ. **g** <TA-MSD> as a function of T for different NaCl concentrations at Φ = 7%. Solid lines in panels **f** and **g** show power-law fits. Data points and solid curves in panels **c**, **e**, **f**, and **g** were vertically shifted for clarity. To distinguish between different symbols associated with time, τ is the lag time, T denotes the temporal interval extending from the first recorded position of a specified trajectory, and T_m represents the sampling time points, excluding the waiting time required for reaching adsorption equilibrium. Source data are provided as a Source Data file.

Fig. 2 ML-based algorithm for classification of diffusion models. a, Workflow of the ML method used to classify diffusion models. Trajectory colors in the dataset creation panel correspond to different diffusion models: ATTM (orange), SBM (cyan), FBM (yellow), BW (gray), CTRW (blue), and LW (black). **b**, Precision (blue), recall (orange), and F1-score (gray) across six models, evaluated on simulated, unseen trajectories. **c–g**, Distribution of predicted diffusion models for experimental trajectories at varying Φ values: ATTM (orange), SBM (cyan), FBM (yellow), BW (gray), CTRW (blue), and LW (dark gray). Source data are provided as a Source Data file.

Fig. 3 Analysis of heterogeneity for Φ = 7%. a Upper: Temporal evolution of *x*-direction displacements (ΔX) for a single representative trajectory (orange). Lower: Corresponding short-time diffusion coefficient (D_s) obtained from linear fits of time-averaged MSD (orange), with three operational regions identified by change-point analysis. Light orange areas correspond to the 95% confidence level. **b** Distribution of normalized displacement $P(\Delta X_\delta, \tau)$. The displacements ΔX_δ within each trajectory are normalized by the standard deviation of that particular trajectory (blue symbol) and fitted with a Gaussian function (blue line). Source data are provided as a Source Data file.

Fig. 4 Molecular dynamics simulation results for two-dimensional charged particles in a Lennard-Jones potential (softness parameter α = 100, reduced temperature θ = 0.1) at Φ = 7% (upper row) and Φ = 1% (bottom row). a, b, e, f, Representative snapshots at waiting times T_m = 0 (**a, b**) and 1000000 (**e, f)**; colors represent the number of particles per aggregate (*N*), as indicated by the color bar. **c, g,** Time-dependent percentages of single nanoparticles and clusters extracted from the MD snapshots; the color bar shows the percentage of each cluster type. **d, h** <TA-MSD> *as* a function of τ for trajectories taken at varying T_m. Colors correspond to waiting times according to the color bar. All quantities are given in standard reduced LJ units. Source data are provided as a Source Data file.

Fig. 5 Gel trapping validation. a Schematic of the gel-trapping technique for replicating nanoparticle aggregates at the oil/air–water interface. The yellow-highlighted regions denote the oil phase, blue boxes indicate the aqueous gellan-gum solution and gelled water phase, and the red dashed line represents the particle monolayer at the oil–water interface. **b-d** SEM images of nanoparticle aggregates on the PDMS gel at early, middle and late gellan-gum curing stages, corresponding to different stages of particle self-assembly. **e** Frequency (%) of single nanoparticles and clusters on the PDMS surface, determined from five representative SEM micrographs obtained via the gel-trapping technique at different time points. The color scale from red to blue indicates the relative frequency percentage (0–100%). **f** Box plot showing the temporal evolution of cluster number across early (blue), middle (gray), and late (orange) stages. The median (horizontal line in box), interquartile range (IQR, box height), and 1.5×IQR whiskers are displayed. n = 67, 93, and 159 technical replicates, respectively. Significant difference was observed between stages (one-way ANOVA, $p < 0.05$), indicating that the aggregate distributions are statistically distinguishable across the measured time points. Source data are provided as a Source Data file.

Editorial Summary:

The adsorption of charged nanoparticles at water-oil interfaces constitutes a fundamental phenomenon, but their diffusional behavior is poorly understood. Here, the authors study diffusional aging by single-molecule tracking, revealing anomalous diffusion dynamics and aggregate growth mechanisms.

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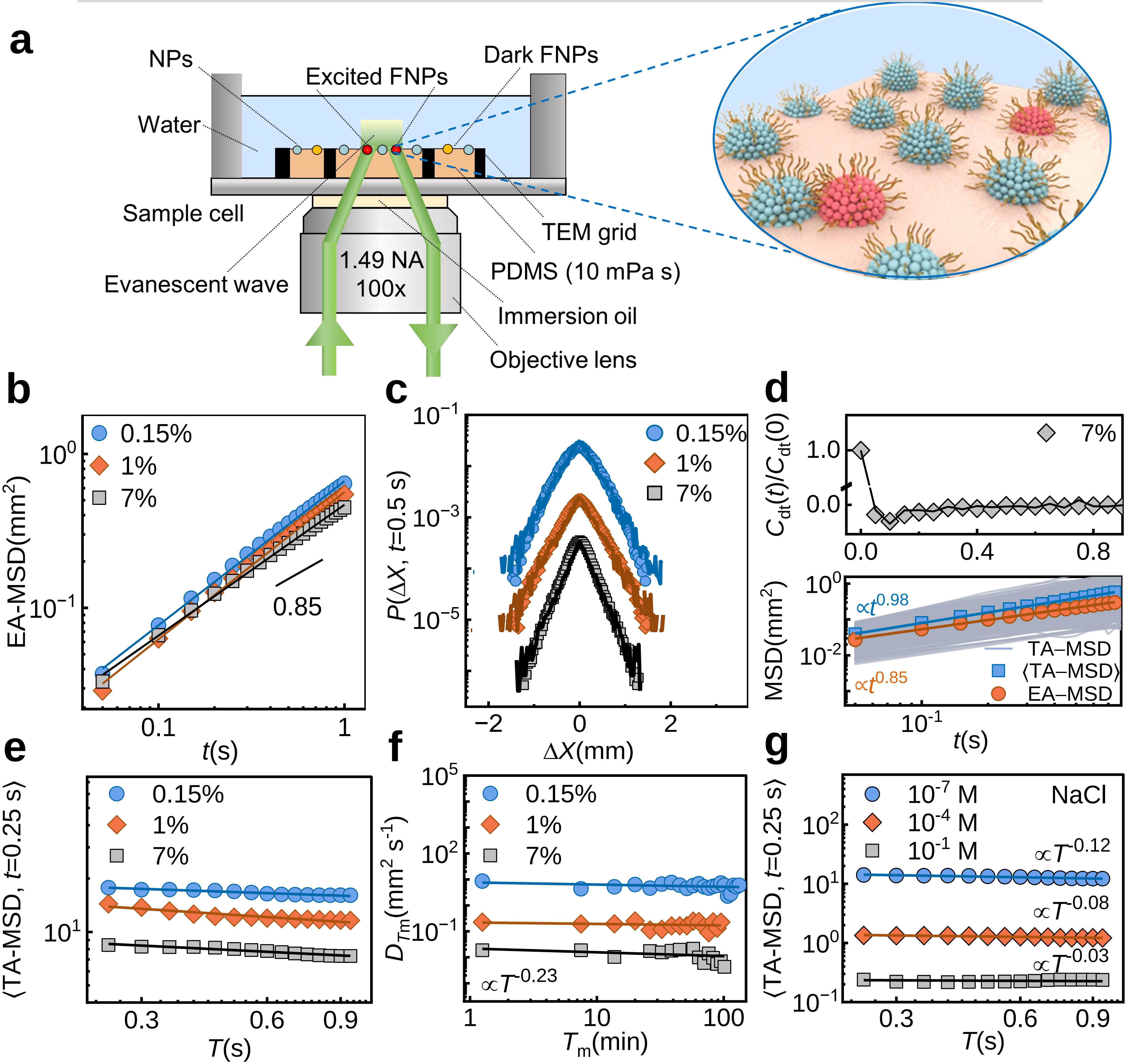
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Objective lens
b
EA-MSD(mm^2)
0.15%
1%
7%
0.85
t(s)
c
$P(\Delta X, t$=0.5 s)
ΔX(mm)
d
$C_{dt}(t)/C_{dt}(0)$
MSD(mm^2)
$\propto t^{0.98}$
$\propto t^{0.85}$
TA–MSD
⟨TA–MSD⟩
EA–MSD
t(s)
e
⟨TA-MSD, t=0.25 s⟩
T(s)
f
D_{T_m}(mm^2 s^{-1})
$\propto T^{-0.23}$
T_m(min)
g
10^{-7} M
10^{-4} M
10^{-1} M
NaCl
$\propto T^{-0.12}$
$\propto T^{-0.08}$
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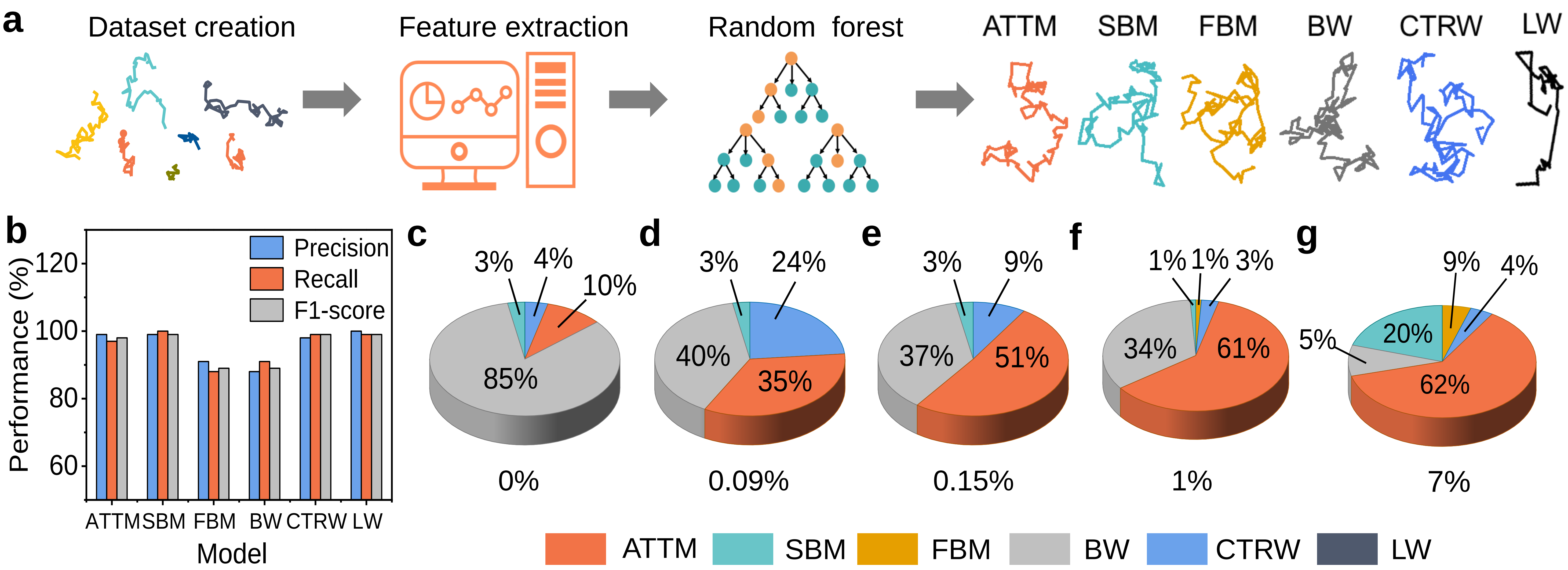
T(s)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Feature extraction
Random forest
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Performance (%)
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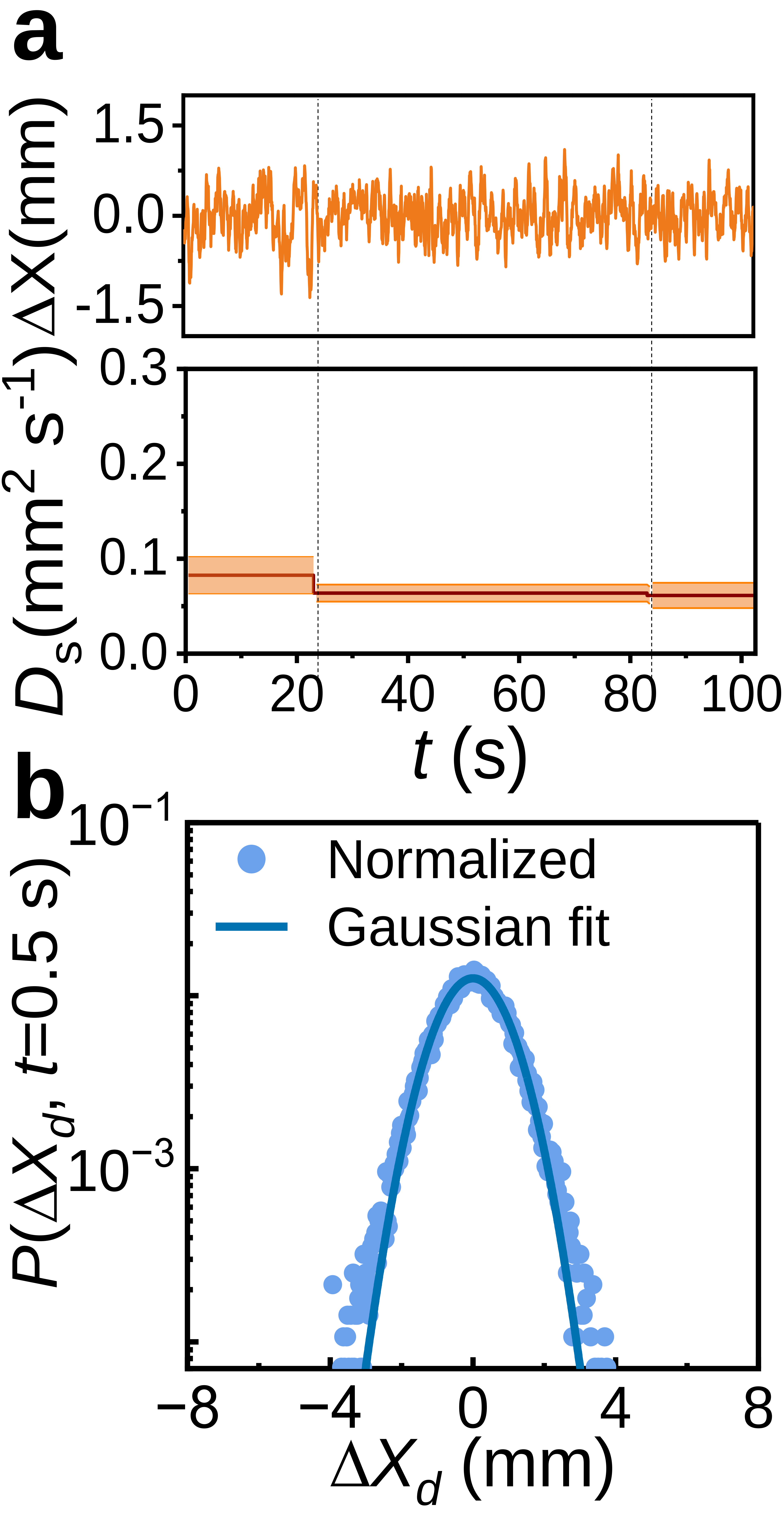
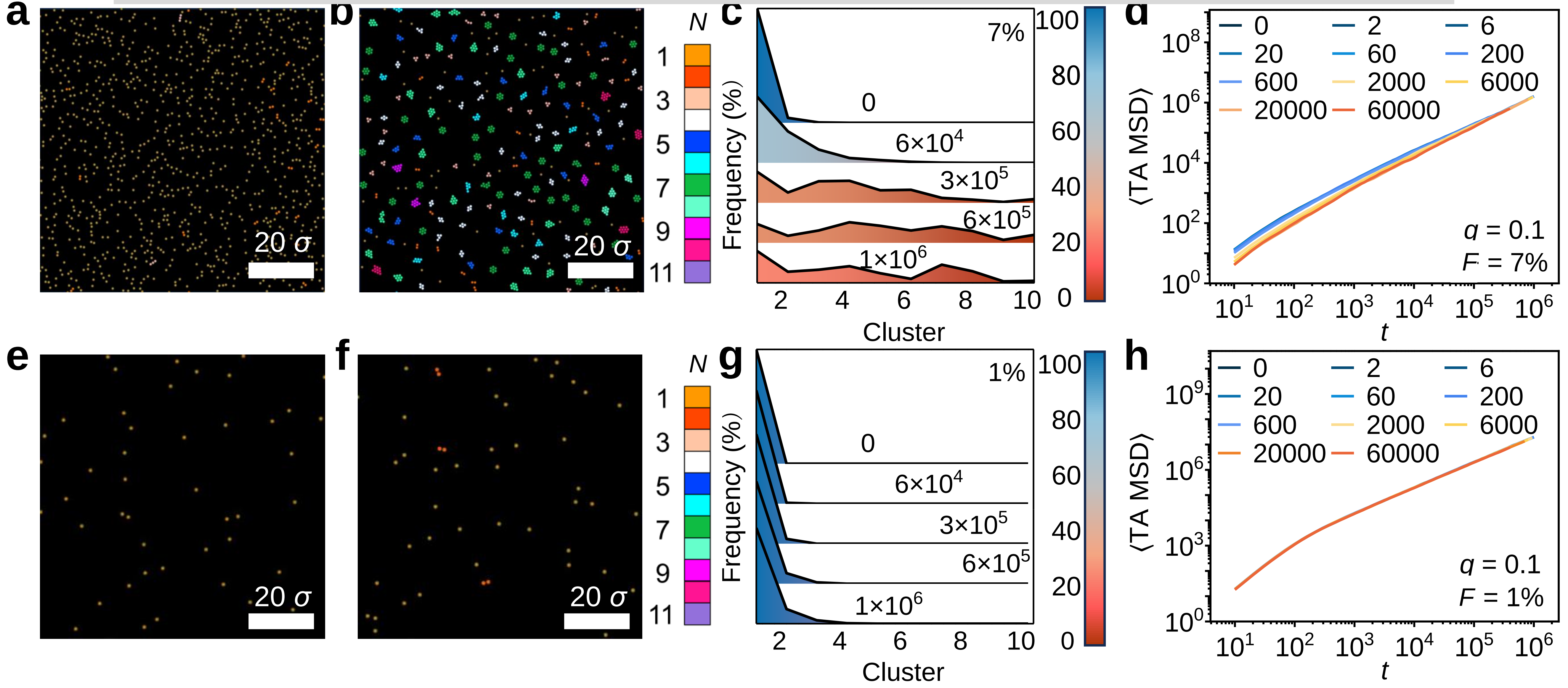

a
ΔX(mm)
1.5
0.0
-1.5
D_s(mm^2 s^{-1})
0.3
0.2
0.1
0.0
0
20
40
60
80
100
t (s)
b
10^{-1}
10^{-3}
P(ΔX_d, t=0.5 s)
Normalized
Gaussian fit
−8
−4
0
4
8
ΔX_d (mm)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Cluster
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⟨TA MSD⟩
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6000
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q = 0.1
F = 7%
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20 σ
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1%
h
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F = 1%


a
50 °C
Oil
Gellan in water
Cooling
25 °C
Oil
Gelled water phase
Remove oil
25 °C
Air
25 °C
Air
PDMS
Pour PDMS
Curing
Peel off
PDMS
Imaging
SEM
2 μm
PDMS
b
2 μm
c
2 μm
d
2 μm
e
Frequency (%)
Early
Middle
Late
Cluster
2
4
6
8
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12
100
80
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40
20
0
f
Number
15
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9
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0
Early
Middle
Late